

Precipitation Reactions in Solution Lab

Purpose: To test the predictability of the solubility table.

Prediction: Use the solubility table to predict the precipitates formed from each of the nine combinations of solutions. Write out the balanced chemical equations for all the reactions. Be sure to include states for all reactants and products.

Procedure:

1. Set up the spot plate so that it matches the above observation table. At the intersection of each row and column, combine **two drops** of the solutions listed.
2. Deposit all waste in waste beaker and clean up all materials.

Observations: Indicate if a precipitate forms from the following reactions. Write yes or no in the table.

	$\text{NaOH}_{(aq)}$	$\text{KI}_{(aq)}$	$\text{Na}_2\text{CO}_{3(aq)}$
$\text{CoCl}_{2(aq)}$			
$\text{AgNO}_{3(aq)}$			
$\text{CuSO}_{4(aq)}$			

Analysis Questions:

1. According to your observations, which chemical reaction conflicts with your prediction? Research this reaction and explain the anomaly.
2. Based on your evidence, which **two** anions (negative ions) from the lab could be used to precipitate most of the metal cations (positive ions)?
3. Why is it necessary to use distilled water rather than tap water to prepare the solutions used in this investigation?
4. Hard water contains a high concentration of calcium ions. Suggest a way to make the hard water softer (what could you add to precipitate out the calcium ions?).
5. Read pages 343-345 (Sequential Qualitative Chemical Analysis) in your text. You are given a solution that may contain either or both phosphate and lead ions. Design a flowchart (use the one on p344 as a guide) to illustrate your testing for the presence of these ions.
6. Write **total ionic** and **net ionic** equations for any 3 reactions from the lab.

